Name: _____

WORKSHEET

Component form of the dot product

Consider the following vectors:

$\mathbf{a} = (2, -6)$	$\mathbf{b} = (-8, 4)$	$\mathbf{c} = (-5, -7)$	$\mathbf{d} = (5, 6)$	$\mathbf{e} = (-4, -1)$	$\mathbf{f} = (-2, 8)$
$\mathbf{g} = 4\mathbf{i} + 3\mathbf{j}$	$\mathbf{h} = 3\mathbf{i} - 5\mathbf{j}$	$\mathbf{k}\,\mathbf{i} - 2\mathbf{j}$	$\mathbf{l} = -7\mathbf{i} + 4\mathbf{j}$	$\mathbf{m} = -3\mathbf{i} - 6\mathbf{j}$	$\mathbf{n} = -6\mathbf{i} + \mathbf{j}$

For the following questions, answer correct to two decimal places where necessary.

1 Write each of the following products in the form $a_1 \times b_1 + a_2 \times b_2$

a $\mathbf{a} \cdot \mathbf{d}$

b $\mathbf{e} \cdot \mathbf{c}$

c $\mathbf{f} \cdot \mathbf{b}$

d $\mathbf{m} \cdot \mathbf{h}$

e $\mathbf{g} \cdot \mathbf{l}$

f $\mathbf{k} \cdot \mathbf{n}$

2 Evaluate the products in question **1**.

a

b

c

d

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e

f

3 Evaluate the following products.

a $\mathbf{c} \cdot \mathbf{f}$

b $\mathbf{h} \cdot \mathbf{l}$

c $\mathbf{b} \cdot \mathbf{a}$

d $\mathbf{d} \cdot \mathbf{e}$

e $\mathbf{k} \cdot \mathbf{g}$

f $\mathbf{n} \cdot \mathbf{m}$

4 Calculate the angle in degrees between the following pairs of vectors, correct to two decimal places.

a **a** and **b**

b **e** and **c**

c **h** and **n**

d **d** and **f**

e **m** and **k**

f **l** and **g**

g **d** and **a**

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h **n** and **g**

i **b** and **c**

j **l** and **m**

k **e** and **f**

l **h** and **k**

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Answers

1 **a** $2 \times 5 + -6 \times 6$

b $-4 \times -5 + -1 \times -7$

c $-2 \times -8 + 8 \times 4$

d $-3 \times 3 + -6 \times -5$

e $4 \times -7 + 3 \times 4$

f $1 \times -6 + -2 \times 1$

2 **a** -2

b 27

c 48

d 21

e -14

f -8

3 **a** -46

b -41

c -40

d -26

e -2

f 12

4 **a** $135°$

b $40.43°$

c $130.43°$

d $53.84°$

e $53.13°$

f $113.39°$

g $121.76°$

h $133.39°$

i $81.03°$

j $93.18°$

k $90°$

l $4.40°$

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